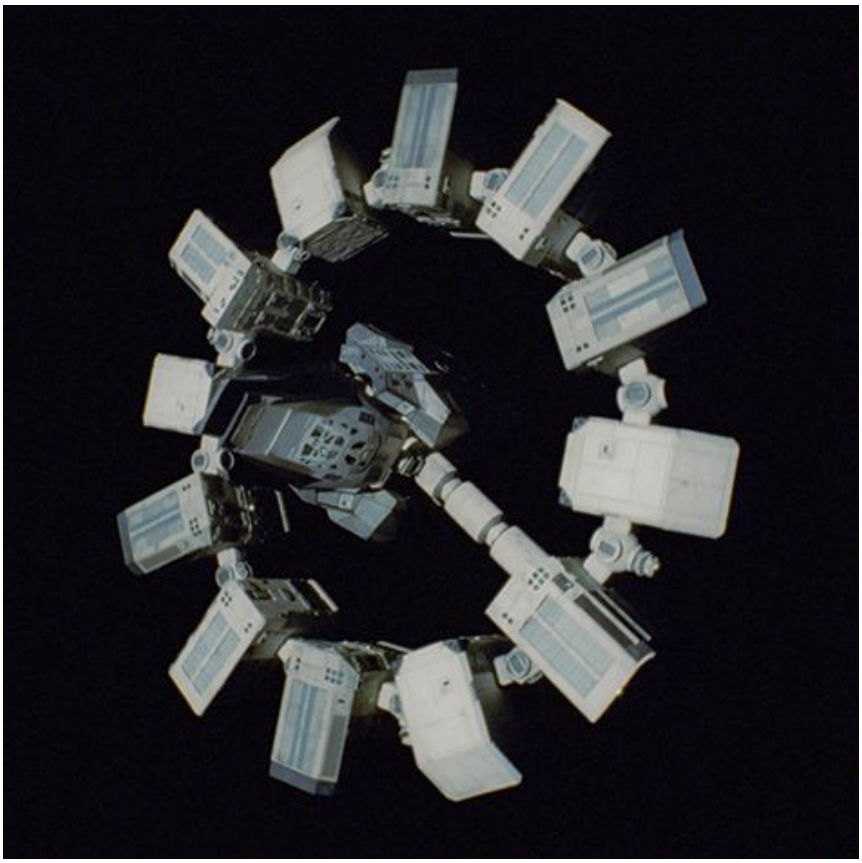


Fig. 20.1. The *Endurance*, with two Rangers and two landers docked onto its central, control module. The Rangers are oriented out of the *Endurance*'s ring plane; the landers, parallel to it. *[From* Interstellar*, used courtesy of Warner Bros. Entertainment Inc.]*

In my scientist's interpretation of the movie, the *Endurance* was designed to survive strong tidal gravitational forces. This was important for the *Endurance*'s trip through the wormhole. The *Endurance* ring's diameter of 64 meters is nearly 1 percent of the wormhole's circumference. Steel and other solid materials break or flow, when subjected to distortions bigger than about a few tenths of a percent, so the dangers were obvious. And little was known about what the *Endurance* would encounter on the Gargantua side of the wormhole, so it was designed to withstand tidal forces far stronger than the wormhole's.